From Local Features To Global Context

Context-Aware Emotion Recognition With Vision Transformers

Simon Reichel[1]

Abstract

Existing visual emotion recognition deep learning models predominantly rely on Convolutional Neural Networks. While these models are very efficient and fast they lack self-attention and global context, often making them reliant on local features with little regard to surrounding scenery. The transformer architecture solves this problem by using multi-head self-attention to gain global context. However, training vision transformer models remains a challenging task, limiting their adoption in social science. We introduce two vision transformer models for emotion recognition by applying supervised transfer learning on pre-trained DeiT-Base and SWIN-Base models. Using the CAER-S dataset (N = 60,000) we successfully predict four emotion classes (Optimistic, Pessimistic, Hostile, Neutral) with strong classification performance ($F1_{SWIN}$ = .829, $F1_{DeiT}$ = .804). Furthermore, we conduct domain adaptation to climate change imagery using a manually annotated subset from the KLIMAMEMES dataset (n = 410) with mixed results ($F1_{SWIN}$ = .759, $F1_{DeiT}$ = .599). Results indicate that SWIN-Base's hierarchical architecture significantly outperforms DeiT-Base's global architecture on small datasets. Even on large datasets SWIN-Base's architecture is better suited for emotion recognition tasks.

Keywords

Emotion Recognition, Computer Vision, Vision Transformer, Transfer Learning

1 Introduction

Visual emotional cues in humans have been studied for centuries (see Bulwer, 1649; da Vinci, 1651) and visual emotion recognition has been an important methodology in social science for decades, offering systematic approaches to studying affective content in visual media and political communication (Russell, 1994; Corneanu et al., 2016). Computational social science tries to implement such methods at large scales which would be prohibitively time-consuming and expensive using manual annotations alone. At the same time the stellar rise of machine learning applications in recent years made the classification of large image corpora easier than ever. Architectures like the vision transformer by Google (Dosovitskiy et al., 2021) have significantly advanced the field of image classifi-

1 Simon Reichel, Ludwig-Maximilians-University Munich, Institute for Communication Science and Media Research, s.reichel@campus.lmu.de

cation, surpassing more traditional methods like convolutional neural networks (CNN) in many areas.

Despite these advances, the adoption of transformer-based architectures in social science research remains limited. Existing computational studies of visual emotion predominantly rely on CNNs (Canal et al., 2022). In their systematic review Pereira et al. (2024) find that more than eighty percent of all papers using deep learning methods for visual emotion recognition relied on CNNs.

On Huggingface (2026), a major repository provider for open-source deep learning models, nearly three million models are currently available. Filtering for image classification reduces this to 27,633 models, and searching for emotion recognition yields 280 results[2], many of which use the transformer architecture. Therefore, it seems like this area of image classification is well covered already. However, on closer inspection, it becomes obvious that many of those models fail to meet basic scientific standards. More often than not, models are poorly documented and provide little information on how they were trained or what training data was used. In many cases, no inference code is provided. Furthermore, judging model performance is difficult as often no metrics are published and even if metrics are available it remains unclear how exactly they were computed. This evidence is only anecdotal, of course. Still, it shows an apparent lack of transformer-based open-source image classification models suitable for scientific application.

In this paper we introduce two novel vision transformer models for emotion recognition which are fully open-source and well documented at the same time. We publish not only model checkpoints but also our entire code from pre-processing to training to inference. Furthermore, all relevant metrics to evaluate model performance are published. We are going to demonstrate model performance as well as applicability by classifying visual media on the topic of climate change. Adaptations to many other domains are possible. We also provide a glossary in the appendix, explaining some of the concepts appearing in this paper.

Our goal is to provide social scientists with models that are both trustworthy and easy to use. Our models completely run on medium-performance local hardware and even on mobile devices, guaranteeing data privacy, independence from third-party infrastructure and cost efficiency. Our entire code is published under a permissive MIT license. Model checkpoints are published under an Apache-2.0 license, granting researchers legal permission to use them free of charge.

2 Last checked: 19 July 2026.

A simplified inference script which allows inference within ten lines of code is published on our Huggingface repository. Therefore, researchers can get our models running within minutes without complicated setup processes. Proficiency in the Python programming language is not required. The models can also be executed from command line without any graphical user interface.

Recognizing our responsibility for the environment now and for future generations our workstation draws all its electrical power from regenerative sources only. All waste heat is fully recycled. We use a closed-loop system for watercooling. These steps allow us to keep environmental impact as low as possible.

2 Related Work

In this section we give a brief overview about the state and taxonomies in classification of emotion in communication science and especially climate change communication so far. However, we dedicate a larger part of this section to explaining the basic technical concepts behind major machine learning architectures, namely convolutional neural networks as well as the vision transformer. As our approach of building a vision transformer for emotion recognition is somewhat novel for social science it is important to understand the advantages our approach brings with it, especially concerning a concept which we call *whole-scene recognition*.

2.1 Emotion classification in visual climate change data

Emotion recognition has long been a subject of interest for many disciplines like healthcare, communication science, education and more recently also computer vision (Shan et al., 2009; Ibrahim et al., 2026; Rahman et al., 2026).

Several taxonomies on how to classify emotion have been proposed. Plutchik (1980) defines 24 emotions organised around eight primary emotions (*joy, trust, fear, surprise, sadness, disgust, anger, anticipation*). Plutchik also defines combinations of emotions like *love*, which is the combination of joy and trust or *optimism* which is the combination of anticipation and joy. Ekman (1992), building on substantial prior work, introduces six base emotions (*happy, sad, anger, fear, disgust, surprise*) as well as a *neutral* class serving as baseline. Ekmans approach has since been used in many datasets and research connected to computer vision and other classification methods (Goodfellow et al., 2013; Mollahosseini et al., 2017; Lee et al., 2019; Duong et al., 2025). A more recent approach by Cao et al. (2014) argues that *surprise* as an emotion differs from the others because it often is more subtle and can occur in combination with any other emotion. In their CREMA-D dataset they propose a reduced taxonomy which omits *surprise* but keeps *neutral* as a separate class (Cao et al., 2014). These taxonomies have since been used in communication science and can be used to study emotions in climate change communication. Climate change messages often seek to raise emotional responses that shape public engagement and support for climate policies (Moser, 2016; Chapman et al., 2017; Brosch, 2021). Generally, climate change communication heavily relies on fear-inducing imagery and also focusses on loss through climate change (O'Neill & Nicholson-Cole, 2009; O'Neill, 2013; Dezma et al., 2024). Positive imagery trying to instil hope exists but is comparatively rare, although in recent years hope has gained more influence in climate change communication, as has anger (Chapman et al., 2017; Nabi et al., 2018; Brosch, 2021). On the other hand, surprise as an emotion has received little attention (Chapman et al., 2017), strengthening the taxonomy proposed by Cao et al. (2014).

In summary, we can identify three meaningful clusters of emotion relevant for climate change communication based on prior research and taxonomies. These three clusters are *optimistic* or hope, *pessimistic* or fear and *hostility* or anger. As our goal is not a fine-grained psychological distinction but rather building a robust and well generalizing tool for automated classification on the topic of climate change we use these three emotion clusters going forward while keeping a fourth *neutral* class as baseline.

2.2 Convolutional and transformer neural architectures

As this paper is aimed at social scientists and not computer scientists we take some time to explain the differences between two major architectures in computer vision, namely vision transformers as well as their predecessors, convolutional neural networks (CNNs). Understanding these architectural differences is key in evaluating their suitability to conduct emotion recognition tasks.

2.2.1 Convolutional Neural Networks

Modern CNNs work by extracting increasingly abstract features through a sequence of layers. They do so by sliding filters, called kernels, over an input image. Each kernel is used to compute feature maps. This process is known as convolution. Each kernel responds to small, local features like edges or specific shapes. Afterwards a non-linear function is used to enable the network to learn complex representations. Then, so called pooling layers reduce spatial complexity by only keeping the most important information. This process of convolution and pooling is usually repeated several times until a few final feature maps remain. These maps are converted to a single vector in a process called flattening. A vector is essentially an ordered series of numbers. The vector is passed to one or more fully connected layers which will conduct the classification by interpreting the series of numbers. For a more detailed description of how a CNN works, see Goodfellow et al. (2016).

A foundational work in the field of computer vision is the *Neocognitron* model by Fukushima (1980). It introduces local receptive fields and hierarchical feature extractions, both of which would become central concepts in later CNN approaches. Later, the modern CNN architecture was pioneered by Lecun et al. (1998) who introduced many of the technologies still used today. Their architecture, known as LeNet-5, already combines convolutional layers and pooling as described above. Early CNNs were held back by a lack of available compute power. This changed in the early 2000's when Buck et al. (2004) introduced the revolutionary Brook framework for parallel computing which later substantially influenced NVIDIA's CUDA framework, enabling fast GPU-accelerated computing. For many years after, CNNs were one of the backbones of computer vision (Krizhevsky et al., 2012; Simonyan & Zisserman, 2014; He et al., 2016). Even after the introduction of the vision transformer in 2021 by Dosovitskiy et al. social science research continues to often rely on CNNs for visual classification tasks (Pereira et al., 2024).

2.2.2 Vision Transformer

2.2.2.1 Patch embeddings and tensors

The vision transformer introduced by Dosovitskiy et al. (2021) is based on the transformer architecture by Vaswani et al. (2017). It starts by dividing one or several images into a series of fixed-size patches. For example an image with a resolution of 224×224 pixels might be divided into 196 patches with 16×16 pixels each. Each of those patches is projected into a vector, called a patch embedding. Additionally, a positional embedding is computed for each patch embedding, as transformers have no inherent understanding of dimensionality (Vaswani et al., 2017; Dosovitskiy et al., 2021). The combined embeddings are stored in a tensor. A tensor is an advanced mathematical object in linear algebra originally introduced by Bernhard Riemann (1868), although formally described only decades later by Gregorio Ricci-Curbastro and Tullio Levi-Civita (1900). In machine learning, tensors are used to store multidimensional data in an ordered and efficient way. They can be imagined as a multidimensional matrix. For example a tensor with shape [x_1, x_2, x_3] is a three-dimensional matrix with x_1 depth, x_2 rows and x_3 columns.

We visualize the process of creating a tensor with all the embeddings in Figure 1:

1. A batch of N images with three colour channels (RGB) and a resolution of 224×224 pixels can mathematically be imagined as a tensor with shape [N, 3, 224, 224].

2. All images get divided into a fixed number of patches. For example using a patch size of 16×16 pixels an image tensor with shape [N, 3, 224, 224] would be converted into a tensor with shape [N, 196, 3, 16, 16] as $\frac{224}{16} \times \frac{224}{16} = 196$ patches get created per image. In practice, patch creation is often implemented in a slightly different way, specifically by convolutional projection, but the concept remains the same.

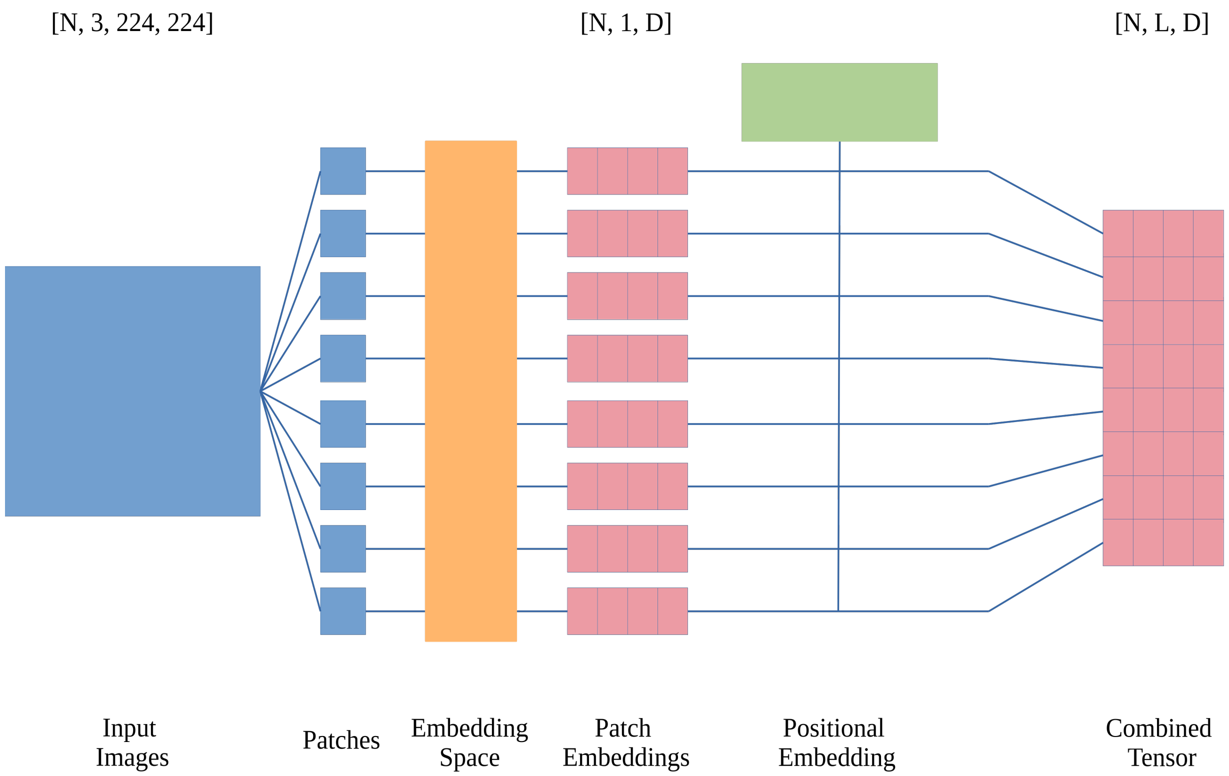
3. For each patch an embedding is computed in an embedding space. Each resulting patch embedding is a tensor with shape [N, 1, D], where N is still the batch-size and D is the number of dimensions. The number of dimensions can differ from model to model.

4. Additionally, a positional embedding for each patch is computed and added to each patch embedding.

5. All patch embeddings together form the final combined tensor with shape [N, L, D] where L is the sequence length. The sequence length is the number of patches. Some models also add a single class token that contains a representation of the entire image to the sequence.

Figure 1

Converting images to patch embedding tensors



Note. N = Number of images (batch-size), D = Number of dimensions, L = Sequence length. Colour codings: Blue: Image, Yellow: Embedding space, Red: Embedding tensors, Green: Positional embedding. Own figure.

2.2.2.2 Encoder-blocks

The tensor is passed through multiple encoder-blocks. Each encoder-block contains multi-head self-attention and a feed-forward neural network. Self-attention as formalized by Vaswani et al. (2017) compares each patch to every other patch, enabling the model to capture both local as well as distant relationships in images. Multi-head attention also allows the model to learn several relationships at the same time. For classification a final tensor is passed to a multi-layer head which conducts classification by interpreting the tensor (Dosovitskiy et al., 2021). All of these processes require a lot of mathematical calculations. For example, the vision transformer by Dosovitskiy et al. (2021) needs to perform around 17.6 billion mathematical operations to classify a single image. More often than not, inference datasets contain thousands of images and training datasets might be even larger, requiring trillions of mathematical operations. Therefore we ask:

RQ1: Can vision transformer models be deployed on local computers without cloud infrastructure?

After having explained the differences in approaching feature extraction and classification we will now explain why the vision transformer is in theory superior to the CNN architecture for visual emotion recognition in the next section.

2.3 Whole-scene recognition

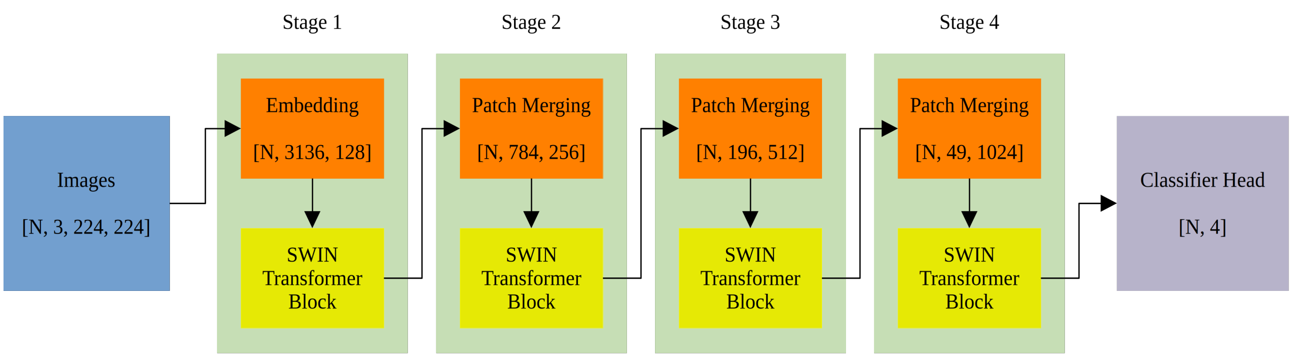
As we explained in section 2.2.1 CNNs extract local features from an image. This makes them an efficient and fast solution for many tasks including facial emotion recognition where local features dominate. However, CNNs are unable to interpret distant features unless they have specialized architectural components. Furthermore, their focus on local features often leaves background scenery somewhat behind. While deep architectures and 3D-CNNs improve awareness, they still lack contextual depth (Mocanu et al., 2023; Wu et al., 2025). Instead of reducing an image through convolution and pooling, a vision transformer keeps all patches of an image together in a tensor. It can then use self-attention to interpret the entire image and gain global context. It is therefore able to learn both local features and long-range relationships. For example, a vision transformer can interpret multiple persons in an image, body posture, background scenery and objects at the same time which is something that most CNNs are unable to do. For some time research has argued that emotion in social context is often complex and cannot be conveyed by facial expressions or other local features only (Barrett et al., 2011; Aminoff et al., 2013; Gao et al., 2015). Instead, interactions between individuals as well as contextual cues in the environment are important indicators for emotional state. Therefore, vision transformers are well suited for in-context emotion recognition. We call this approach *whole-scene recognition* to describe the capability of simultaneous local and global feature interpretation, which distinguishes it from face-only approaches. In summary, we ask:

RQ2: Can vision transformer models translate their theoretical strengths into real-world results and achieve good prediction performance on visual emotion recognition?

2.4 Vision transformer models: SWIN-Base and DeiT-Base

In this paper we are using two pre-trained vision transformer models, specifically *Microsoft SWIN-Base-patch4-window7-224* by Liu et al. (2021) and *Meta DeiT-Base-patch16-224* by Touvron et al. (2021). Both models are pre-trained on the ImageNet-1k dataset and licensed under Apache-2.0. We explained patch embeddings and tensors in section 2.2.2.1 which is important to understand the major difference between the two models. DeiT-Base is close to a pure transformer architecture as described by Dosovitskiy et al. (2021) and will be used as a transformer baseline. It uses patch embeddings with shape [N, 197, 768] and performs self-attention over all patches simultaneously in each encoder-block before returning a classification (Touvron et al., 2021). SWIN-Base on the other hand is a hierarchical vision transformer and uses a shifted window approach with four stages depicted in Figure 2.

Figure 2
SWIN transformer stages



Note. N = batch-size. Integers inside square brackets show tensor shapes. Colour codings: Blue = Images, Orange = Embeddings, Yellow = Transformer blocks, Green = Stages, Violet = Classifier head. Own figure.

It first creates a large number of small patches which are embedded in a 128-dimensional space returning a tensor with shape [N, 3136, 128]. In a machine learning context 128 dimensions are considered low-dimensional compared to DeiT-Base's 768 dimensions. This low-dimensional embedding increases local feature extraction performance. Inside the encoder-blocks self-attention is not performed over all patches simultaneously. Instead patches are divided into small groups, which are called windows. Self-attention is computed for each window individually. In the next encoder-block those windows are shifted, so over time each patch is compared to every other patch. After each stage some of the patches are merged creating larger and larger patches which are embedded in increasingly higher-dimensional spaces. While DeiT-Base keeps the same [N, 197, 768] embedding structure throughout, SWIN-Base changes its embedding structure after each of its first three stages as described in Table 1.

Table 1
Embedding tensors in SWIN-Base

Stage	SWIN-Base tensor shape
Initial patch embedding	[N, 3136, 128]
Stage 1	[N, 3136, 128]
Patch merging	[N, 784, 256]
Stage 2	[N, 784, 256]
Patch merging	[N, 196, 512]
Stage 3	[N, 196, 512]
Patch merging	[N, 49, 1024]
Stage 4	[N, 49, 1024]
Classifier head	[N, 4]

Note. N = batch-size. Classifier head returns a tensor with 4 classes as specified in section 2.1.

In theory, the shifted window approach used by SWIN-Base combines the advantages of the CNN-architecture like efficient local feature extraction with those of the transformer-architecture like global context (Liu et al., 2021).

After having established the theoretical advantages of the transformer architecture and explaining different approaches within this architecture as well as identifying a four-class taxonomy for emotion recognition in climate change communication, we now describe how exactly we implement all of these concepts into two working classification models.

3 Method

In this section we describe how to apply transfer learning as well as domain adaptation to two vision transformer models. Transfer learning is an approach in machine learning in which an existing model pre-trained on a large source dataset is adapted to a new task by further training it on task-specific data. It requires less data and is computationally cheaper compared to training from scratch. We use two datasets, specifically the CAER-S dataset by Lee et al. (2019) for training as well as a licensed subset of the KLIMAMEMES dataset by Haim et al. (2026) for domain adaptation and inference.

3.1 CAER-S dataset

3.1.1 Dataset description

Applying transfer learning to transformer models requires vast amounts of data. We use the CAER-S dataset by Lee et al. (2019). This dataset is a derivative of the original CAER dataset by the same authors. The CAER dataset contains around 13,000 annotated videos of films and TV series. The CAER-S dataset on the other hand consists of 69,999 still images from those videos. Images therefore resemble in-the-wild scenes but contain professional actors in a light-controlled environment and a professional set design as well as professional cinematography. All 69,999 images are annotated by human annotators. Each image is annotated with exactly one out of seven emotions (Anger, Disgust, Fear, Happy, Neutral, Sad, Surprise). Furthermore, Lee et al. already split the dataset into train/test directories (2019). Becuase we conduct pre-processing on the data anyway, we are not following their suggestion and instead create our own splits. The CAER-S dataset is around 13.5 GB in size and is available here: https://caer-dataset.github.io/.

3.1.2 Legal Use

Pursuant to Article 5.3a of Directive 2001/29/EC (InfoSoc Directive), which is part of European Union law, member states of the European Union shall be allowed to make exceptions for the use of copyrighted material for non-commercial research purposes. Germany, in its capacity as a member state of the European Union, has allowed for the following exceptions:

Pursuant to Urheberrechtsgesetz (UrhG) §§ 60c, 60c Abs. 2, 60d researchers shall be eligible to copy, store, process and distribute major parts of copyrighted material for non-commercial research purposes.

Furthermore, pursuant to Directive (EU) 2019/790 (DSM Directive), Articles 3 and 4, member states of the European Union must provide exceptions for text and data mining. Germany, in its capacity as a member state, implemented these provisions primarily through §§ 44b and 60d UrhG. Under § 60d UrhG, research organisations and their members may make copies of lawfully accessible works and process them for scientific text and data mining, including machine-learning training. Rightholders cannot opt-out. The use of the CAER-S dataset in this paper is therefore legitimate under European Union and national law.

3.2 KLIMAMEMES dataset

3.2.1 Dataset description

We want to use our trained classifiers on *real-world* data. With real-world data we mean images which have not been created in an artificial environment like a film set but "in-the-wild". Real-world data usually does not contain professional actors or a professionally controlled background. We still allow professional speakers like politicians to be part of an image. We also allow professional settings like exhibitions. Real-world images may be created by a professional photographer or journalist as long as the image itself is at least somewhat natural and does not contain fictional characters or scenes.

To comply with these requirements we use a subset of the KLIMAMEMES dataset by Haim et al. (2026). According to the codebook by Luebke et al. (2025) the KLIMAMEMES dataset contains social media images containing at least one of 150 keywords or which is published by one of 130 stakeholders. Keywords are hashtags from social media post captions and might reference events like the COP28 or COP29 conference, the fight against climate change, climate scepticism or others. Stakeholders may be politicians, organisations, media outlets and others. Data has been scraped from Facebook, Instagram, TikTok and YouTube from 27.11.2023 to 16.12.2023 as well as from 04.11.2024 to 01.12.2024 (Luebke et al., 2025).

We only use a subset of the dataset. First of all, we only use images from Instagram which captions included at least one of the *climate activists* keywords (N = 8,190). We further reduce complexity by only using images that contain at least one human face. To select these images, we use the *Insightface SCRFD* face detection model inside Python 3.13.13. Full code is provided on our Github (https://github.com/47867/ViT-ER). Of the 8,190 images, 4,450 contain at least one human face. Using only images with humans present is sensible for our goal to classify emotion. Our classifiers have been explicitly trained on images that also contain humans, namely the CAER-S dataset. We therefore should also inference on images that are somewhat similar.

3.2.2 Legal use

We have obtained a license that allows us to copy, store, edit, adapt, build onto, but not distribute certain parts of the dataset for scientific purposes free of charge. The license covers the use of the dataset in training and inferencing with machine learning models. The license has been obtained from Nadezhda Ozornina, M.A., LMU Munich.

3.3 Categories

The CAER-S dataset we are using contains seven emotions identical to those proposed by Ekman (1992). We follow a more recent approach by Cao et al. (2014) and drop *surprise* as an emotion. Surprise is often a subtle emotion that can easily be misclassified as any other emotion (Cao et al., 2014). Furthermore, we are classifying images with a connection to climate change activism. As outlined in section 2.1 optimistic, pessimistic and hostile emotions are well established categories in climate communication research while surprise has received little attention (O'Neill & Nicholson-Cole, 2009; Chapman et al., 2017). For transfer learning we therefore cluster the six remaining emotions to four classes in accordance with prior research. Those classes are *Optimistic* (Happy), *Pessimistic* (Sad and Fear), *Neutral* (Neutral) and *Hostile* (Anger and Disgust). Neutral is used as a baseline in accordance with prior research (Ekman, 1992; Cao et al., 2014). Our codebook for the four described emotion classes is adapted from Luebke et al. (2025) and part of the appendix.

3.4 Transfer learning

We apply supervised transfer learning to both SWIN-Base and DeiT-Base. In this paper transfer learning means we replace the ImageNet-1k classifier head with a custom four-class classifier head and subsequently fine-tune the hidden model layer weights through supervised training. The hidden model layers are usually referred to as *the backbone*. Instead of training from scratch, we leverage the knowledge the models have already gained during training on the ImageNet-1k dataset. This approach is computationally cheaper and requires less, but still substantial amounts of data. In simple terms, we use a model that already exists and fine-tune it to our specific task. We validate both models over validation loss, accuracy, precision, recall and unweighted F1-scores. Validation results are presented in the results section.

3.4.1 Data pre-processing

Before images can be used in training they need to undergo some transformations. First of all, we employ augmentations. Augmentations change certain aspects of an image in a pseudo-randomized way. They therefore increase the variety of samples a model sees each epoch. An epoch is one complete pass of the training dataset through the model. We use *HorizontalFlip*, *RandomCrop*, *Brightness*, *Contrast*, *HSV*, *Rotate*, *Gaussian Noise* and *Coarse Dropout*, all with $0.3 \leq p \leq 0.5$. Those augmentations are only applied to the training images. Validation images are not augmented. Additionally, our two transformer models expect images that have been normalized using the ImageNet standard (Mean: (0.485, 0.456, 0.406); SD: (0.229, 0.224, 0.225)). Images are resized to 224×224 pixels and converted to tensors of shape [3, 224, 224], where 3 indicates an RGB-image with three colour channels and 224 is the width and height of the image. We use a stratified 80/20 split over all tensors for train and test splits.

Our classes are imbalanced. Pessimistic and Hostile each make up 33% of the dataset, while Optimistic and Neutral only make up around 17% each. To account for imbalance, a weighted random sampler is used which oversamples minority classes to produce approximately class-balanced training batches. As we have chosen a batch size of 64, our final image tensors have a shape of [64, 3, 224, 224]. Class weights are computed as inverse square-root frequencies to further address class imbalance. During training class weights are passed to the criterion and influence the loss function. All weights are computed with double precision (FP64).

3.4.2 Training

Both transformer models are trained on the exact same training process with four stages. We use a mixed precision approach to reduce compute cost but keep precision high where it matters most. All model weights are stored with FP32 precision. Gradient scaling as well as optimiser steps are also conducted with FP32 precision. Only forward passing and loss computation run with lower FP16 precision. Gradients are unscaled to FP32 before the optimiser step. Image tensors with shape [64, 3, 224, 224] are converted to embeddings with shape [64, 197, 768] for DeiT-Base and [64, 3136, 128] for the first stage of SWIN-Base.

In Stage 1 only the classifier head is trained while the backbone remains frozen to allow the head to learn from stable features. For loss computation we use weighted cross-entropy loss, which uses the class weights described in the pre-processing section. After three epochs, we enter Stage 2a and enable MixUp ($\alpha = 0.2$) as specified by Zhang et al. (2018). MixUp blends pairs from two samples using a mixing coefficient sampled from a beta distribution and creates a soft label for the blended sample. This process improves both model generalization and robustness. The backbone remains frozen for two more epochs to stabilise training.

With five epochs finished the training enters Stage 2b. The backbone is unfrozen and remains that way until the end of training. Stochastic depth is also activated ($droppath = 0.1$), which randomly drops transformer blocks during training to regularise the backbone. As epoch six is the first epoch with an unfrozen backbone we disable MixUp for this epoch again to stabilise training before introducing more complexity. In epoch seven MixUp is re-enabled with the same settings as before. The training configuration remains fixed for all subsequent epochs. A cosine annealing learning rate scheduler is used throughout training. For stages 1 and 2a the head learning rate is set to 1×10^{-4}. At the start of Stage 2b, head learning rate is reduced (2×10^{-5}) and a 3-epoch linear warmup is applied before cosine annealing resumes. The backbone is trained at a lower learning rate (5×10^{-7}) than the classification head to prevent catastrophic forgetting of pretrained features. To prevent overfitting we use a drop rate of 0.3.

Training is conducted for a maximum of 35 epochs. If validation loss has not reached a new global minimum for seven consecutive epochs training is stopped early. The checkpoint with the lowest validation loss is kept in memory and saved at the end of training. For compatibility reasons checkpoints are first stored as .pth files and later converted to the more modern .safetensors file format. We recommend to use only .safetensors files. The .pth file format poses a potential security risk and its use is discouraged.

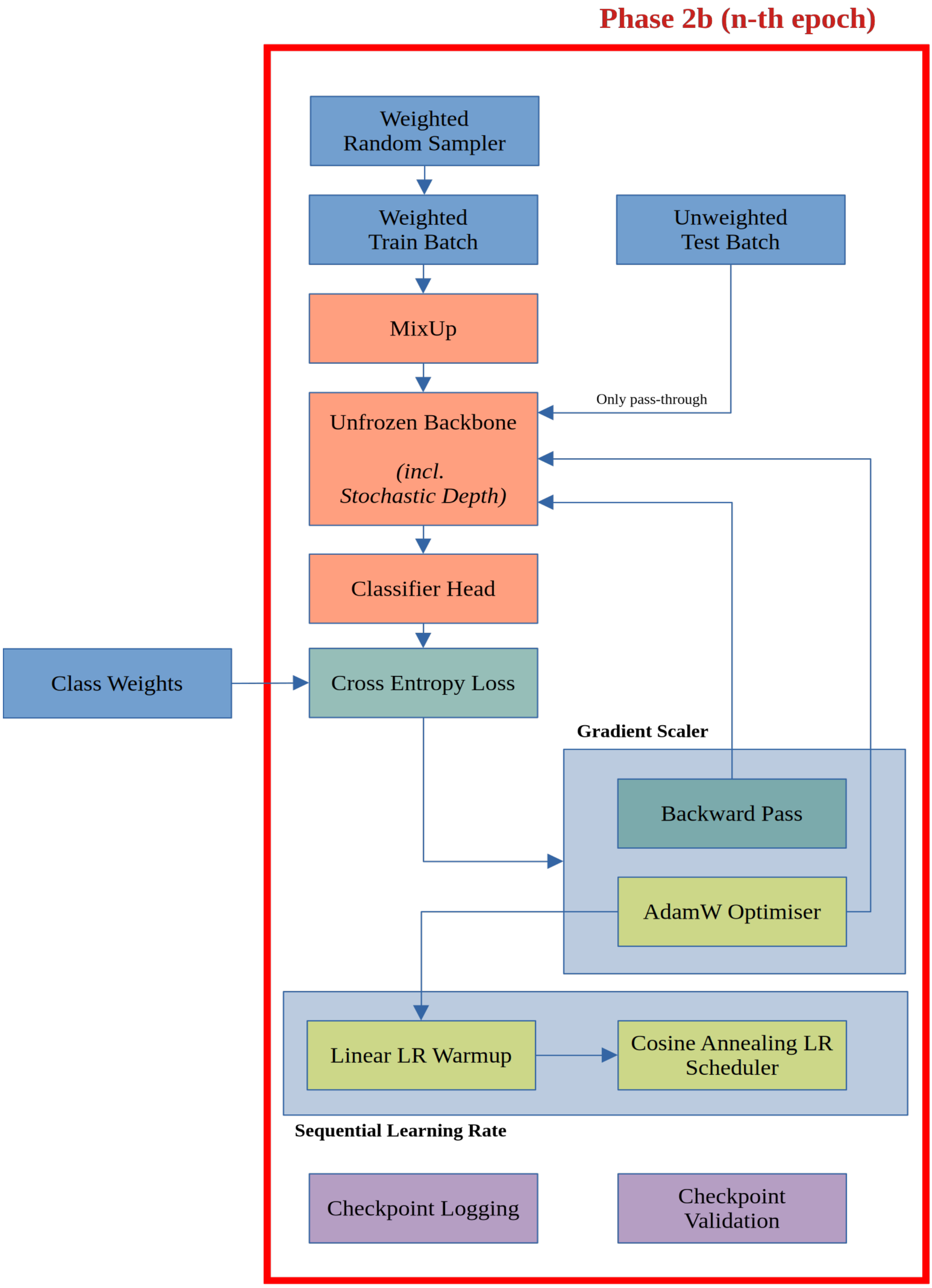
All four stages are summarized in Table 2. A graphical overview of the training process is supplied in the appendix with a visualization of stage 2b n-th epoch in Figure 3.

Table 2
Training process (4 stages)

Stage	Epoch	Description
Stage 1	Epochs 1-3	Head only, backbone frozen
Stage 2a	Epochs 4-5	Head only, MixUp enabled, backbone frozen
Stage 2b (first epoch)	Epoch 6	Backbone unfrozen, stochastic depth enabled, MixUp disabled, linear lr-warmup (3 epochs)
Stage 2b (n-th epoch)	Epochs 7-35	Backbone unfrozen, stochastic depth enabled, MixUp re-enabled

Note. Full training process applied to SWIN-Base and DeiT-Base each. Training stopped early after 32 epochs each after validation loss reached no new minimum for seven consecutive epochs.

Figure 3
Training Stage 2b (n-th epoch)



Note. Part of own figure. Full figure in the appendix. Figure describes the process of one epoch *n* in the training process with $6 < n < 36$. Class weights are computed at pre-processing stage (not shown here) and are therefore outside the epoch block (red marked). Colour codings: Blue = Data, Orange = Model, Green = Loss computation, Yellow = Optimiser and learning rate, Purple = Validation.

3.5 Domain Adaptation

After training and validating (see section 4) our model checkpoints we move on to domain adaptation. Our checkpoints are trained on film stills for emotion recognition. Expression of emotion in films might differ from real world data. Films are well lit, make use of professional set design and rely on professional actors to bring certain emotional states across. While using data like this for transfer learning is a good starting point, models trained this way can have difficulty with classifying real world data where emotion might be more confusing and subtle.

To improve model predictions we need to conduct domain adaptation. Domain adaptation is a process in training which aligns the model with a certain type of data. This enables us to shift the model to make good predictions on images that significantly differ from the general training data. Technologically, we conduct more training similar to the general transfer learning described in section 3.4. However, this time we do not replace the classifier head, as it already classifies the four emotion classes we need. Furthermore, only a subset of the backbone is unfrozen to prevent overfitting on our much smaller sample compared to general transfer learning. Again we validate both models over validation loss, accuracy, precision, recall and unweighted F1-scores. Validation results are presented in the results section.

3.5.1 Data pre-processing

To conduct supervised domain adaptation, we manually annotate a subset of our real-world data (n = 410). A stratified 80/20 split is used for training and validation data. As training data is very limited we employ aggressive augmentation to artificially increase the number of samples the models see in each epoch (HorizontalFlip, RandomCrop, Brightness, Contrast, HSV, Rotate, GaussianNoise, CoarseDropout, ColorJitter). Only images in the training split are augmented.

Both training and test images are normalized to ImageNet standards and resized to 224×224 pixels. We are choosing a smaller batch-size of 16 compared to general transfer learning batch-size of 64. This is necessary as our domain adaptation dataset is much smaller. The number of batches must not be too small to allow for sufficient gradient updates therefore requiring to make the individual batches smaller. Like in general transfer learning we employ several weighting techniques to account for class imbalances. We use a weighted random sampler to build our train batches which consist of tensors with shape [16, 3, 224, 224]. The weighted random sampler ensures that each batch has roughly the same amount of images per class to improve training stability. We also calculate class weights as inverse square-root frequencies over our class distribution. These weights are passed to the criterion and influence the loss function. Finally, to improve accuracy, we assign sample weights to images that were consistently misclassified during preliminary training runs, identified by manual inspection, to increase their influence on the loss function.

3.5.2 Training

Using the annotated data we conduct further supervised training with our model checkpoints. We use the same mixed-precision approach as described in the general training section and train for 80 epochs. If validation loss does not reach a new global minimum for 15 consecutive epochs training is stopped early. Only the classifier head and around a third of all backbone paramteres are unfrozen (*layers.3* for SWIN-Base, *layers.7* through *layers.11* for DeiT-Base) to prevent overfitting. Both head and backbone learning rates are conservative to stabilise training ($lr_{head} = 5 \times 10^{-5}$, $lr_{backbone} = 5 \times 10^{-6}$). Furthermore we implement MixUp ($\alpha = 0.125$) to improve generalization over the data. The checkpoint with the lowest validation loss is kept in memory and saved at the end of training as a .safetensors file.

Our entire code as well as manual annotations are open-source and published on our Github under a permissive MIT-license: https://github.com/47867/ViT-ER. Model checkpoints and simplified inference code are published on Huggingface under an Apache-2.0 license: https://huggingface.co/Simon14142/emotion-recognition-4class-swin (replace *swin* with *deit* in URL for second model). SHA256 checksums are provided for file integrity verification.

3.6 Inference

The checkpoint from training with the lowest validation loss is used for inference for both SWIN-Base and DeiT-Base. All model weights are stored in FP32 precision. Inference is possible with single-precision (FP32) or with half-precision (BF16) to save compute cost while sacrificing some accuracy. We recommend BF16 over FP16 as BF16 shares the same 8-bit exponent used by FP32, minimizing the risk of numerical overflow. All inference results in this paper are computed with FP32 precision as we have enough compute power available.

We run inference on the 4,450 images described in section 3.2.1. All images are resized to 224×224 pixels, normalized to ImageNet standards and transformed into a tensor. The dataloader batches 64 tensors together to a final tensor with shape [64, 3, 224, 224] which is then processed by the hardware accelerator. Results are stored in a *pandas* dataframe and later exported to a UTF-8 encoded .csv file. Results contain seven columns: The image path, the final prediction out of the four classes, a confidence score which is computed as a 32-bit float and additionally confidence scores for each class which are also computed as 32-bit floats. If confidence for all four classes is below a pre-determined threshold (*t = 0.4*), no prediction is made and instead the label *Unknown* is cast.

3.7 Hardware and requirements

Transfer learning, domain adaptation and inference are all conducted on local hardware. Data privacy and 100% control over all processes are thus guaranteed at all times. All electrical power used for training and inference stems from renewable energy sources. Waste heat is 100% recycled. We therefore are able to keep environmental impact as low as possible. Our workstation uses a NVIDIA Blackwell GB203 accelerator with 16 GB (~ 16,303 MiB) of GDDR7 VRAM (28 Gigabit/s over 256-bit bus = 896 GB/s transfer speed (~ 834 GiB/s), non-ECC) and an Intel 12th. Gen Core i5-12600K with 16 logical cores and a boost clock of 4.90 GHz. GPU and CPU are directly connected using a PCI-Express Gen. 5.0 interface with 16 lanes. As mass storage a 1 TB (~ 928.2 GiB) SAMSUNG 870QVO solid state drive is used which is connected via SATA-III-600 over the CPU chipset (Intel Z690 for LGA-1700). Additionally, the CPU has access to a total of 64 GB DRAM running at JEDEC specification DDR4-3200 MT/s at 1.35V and 16-20-20-38 timings. Additional hardware information are part of the appendix. Software-side we are using Linux Fedora 43 Workstation and the Linux kernel 7.0.12-101.fc43.x86_64. The GPU uses NVIDIA CUDA 12.8 as well as the NVIDIA driver 580.159.04. The entire code runs in JupyterLab inside a conda environment with Python 3.13.13. A *requirements.txt* is published on our Github.

At batch-size of 64, training takes around 1 hour for DeiT-Base (approx. 7 batches per second) and 1.5 hours for SWIN-Base (approx. 5.2 batches per second). DeiT-Base needs approx. 6.0 GiB of VRAM while SWIN-Base needs approx. 11.4 GiB of VRAM. Training should therefore be possible on a GPU with 12 GiB of VRAM. We still recommend a GPU with at least 16 GiB of VRAM. DRAM usage stays below 10 GiB for both models. Therefore, 16 GiB of DRAM should suffice, but 24 GiB or more are recommended.

4 Results

4.1 Model checkpoint validation

Table 3
Performance metrics for model checkpoints

Class	Metric	CAER-S		Domain-Adapted	
		SWIN-Base	DeiT-Base	SWIN-Base	DeiT-Base
Optimistic	Precision	.820	.795	.857	.643
	Recall	.845	.810	.720	.720
	F1	.832	.802	.783	.679
Pessimistic	Precision	.925	.906	.583	.375
	Recall	.869	.864	.700	.600
	F1	.896	.884	.636	.462
Hostile	Precision	.882	.882	.833	.727
	Recall	.853	.802	.769	.615
	F1	.867	.840	.800	.667
Neutral	Precision	.671	.612	.784	.667
	Recall	.778	.776	.853	.529
	F1	.721	.689	.817	.590
Macro-F1		.829	.804	.759	.599
Accuracy		.844	.820	.780	.610
Validation Loss		.429	.492	.608	.794
Cohen’s Kappa		---	---	.668	.454

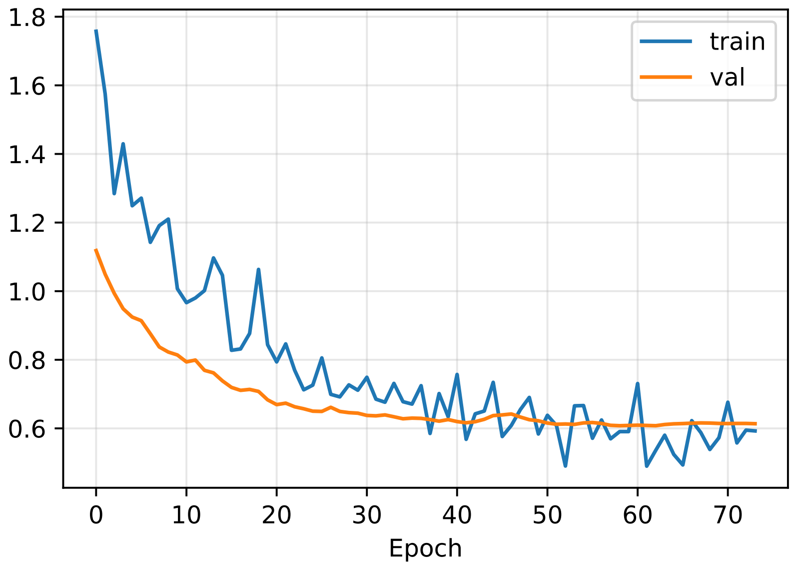
Note. All metrics are unweighted. CAER-S models evaluated on the CAER-S validation set (N = 12,000). Domain-adapted models evaluated on the manually annotated KLIMAMEMES subset (N = 82). Cohen’s Kappa only computed for domain-adapted models as comparison to human gold standard.

RQ1 asks if vision transformer models can translate their theoretical strengths to real-world applications like emotion recognition. To validate model checkpoints and ultimately answer *RQ1* we use validation loss, accuracy, precision, recall and unweighted F1-scores. In initial transfer learning both models achieve strong results on the CAER-S validation dataset. Both models stopped training early after 32 epochs. SWIN-Base reaches a unweighted macro-F1 of 0.829 (accuracy = .844, validation loss = .429). DeiT-Base on the other hand is slightly weaker wit ha macro-F1 of 0.804 (accuracy = .820, validation loss = .492). Both models deliver the best results on *Pessimistic* ($F1_{SWIN}$ = .896, $F1_{DeiT}$ = .884) and *Hostile* ($F1_{SWIN}$ = .867, $F1_{DeiT}$ = .884). The weakest class is *Neutral* for both models ($F1_{SWIN}$ = .721, $F1_{DeiT}$ = .689). SWIN-Base consistently outperforms DeiT-Base across all metrics. Without domain-adaptation both models show very low accuracy on our real-world data from the KLIMAMEMES dataset. SWIN-Base still has the edge (F1 = .292) over DeiT-Base (F1 = .259) but both models are not usable.

After domain-adaptation, both models significantly increase their performance on the KLIMAMEMES dataset but, as expected, remain weaker compared to the CAER-S validation data. SWIN-Base stopped training early after 74 epochs and managed an unweighted macro-F1 of .759 (accuracy = .780, validation loss = .608). Training curves show a plateau in accuracy and F1 after around 40 epochs. Loss curves indicate no further improvement but also no overfitting. DeiT-Base trained for the full 80 epochs but is significantly weaker with a macro-F1 of .599 (accuracy = .610, validation loss = .794). Accuracy and F1 scores plateau after around 60 epochs. The loss curves indicate no overfitting. SWIN-Base classifies *Neutral* (F1 = .817), *Hostile* (F1 = .800) and *Optimistic* (F1 = 0.783) accurate, but is weaker with the *Pessimistic* class (F1 = .636). DeiT-Base struggles across all classes. *Pessimistic* is the weakest class (F1 = .462), while *Optimistic* is the strongest class (F1 = .679). Cohen's Kappa as inter-coder measurement between human and machine is .668 for SWIN-Base and .454 for DeiT-Base. Following Landis and Koch (1977) this indicates moderate agreement for both models, although SWIN-Base is close to substantial agreement. In summary, we have mixed support for *RQ1*. On the one hand, both models deliver strong performance in general transfer learning. However, only SWIN-Base manages to generalize to a new dataset.

Figure 4

Loss curves for domain-adapted SWIN-Base



Note. During domain-adaptation SWIN-Base trained for 74 epochs and then stopped early. Only the head and around a third of all backbone parameters were unfrozen to prevent overfitting. Train/test splits were created by using a stratified 80/20 split (N = 410). Spikes in training loss curve due to small batch size (16). Figures for other model checkpoints in the appendix. Own figure.

4.2 Inference performance

Our *RQ1* asks if vision transformer models can run on local computers without cloud infrastructure. We therefore measure how much time our accelerator needs to classify all 4,450 images. The NVIDIA Blackwell GB203 delivers around 44 trillion mathematical operations per second in FP32 precision with batch-size of 64 and processes the entire dataset in eleven seconds (378 images per second). We also inference the dataset on a much slower everyday mobile device, specifically an Apple MacBook Air M4 for comparison. The MacBook Air needs around 143 seconds to process all 4,450 images (31 images per second) at FP32 precision, delivering around three trillion mathematical operations per second. We can therefore answer *RQ2*: Transformer models can run on local workstations. They can even run on mobile devices, although significantly slower of course. Cloud infrastructure is not required.

Table 4

Inference performance

	Single precision (FP32)	Half precision (BF16)
NVIDIA Blackwell GB203	378 img/s	755 img/s
Apple MacBook Air M4	31 img/s	---[a)]

Note. N = 4,450. Batch-size = 64. img/s is an abbreviation for images per second. Apple MacBook Air M4 with 10 CPU cores, 8 GPU cores, 16 neural engine cores, 16 GB DRAM, 120 GB/s transfer speed. NVIDIA Blackwell GB203 uses NVIDIA CUDA for GPU acceleration. Apple MacBook Air uses Metal Performance Shaders (MPS) for GPU acceleration.

a) MacBook Air does not support BF16 precision natively on GPU or neural engine.

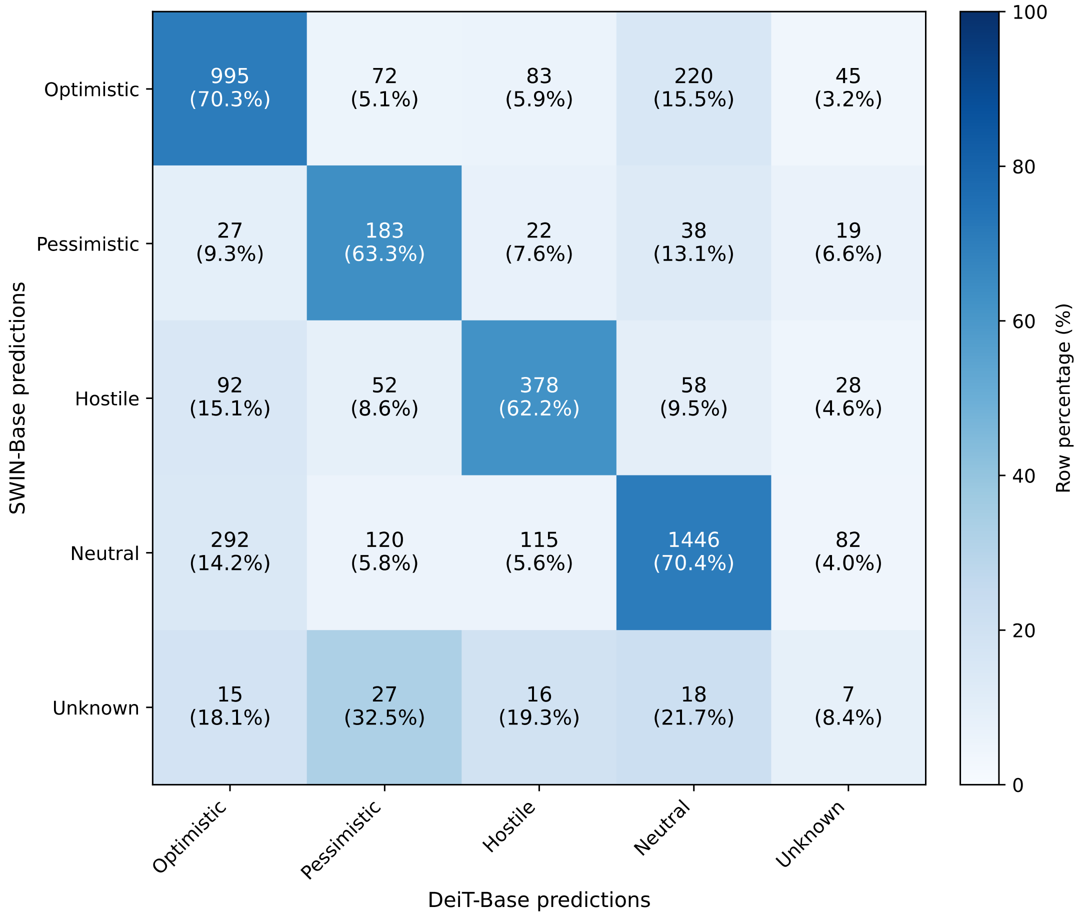
4.3 KLIMAMEMES results

Finally, we want to take a short look on our inference data from the KLIMAMEMES dataset. As outlined in the method section all results are computed with FP32 precision on the NVIDIA Blackwell GB203 accelerator. If prediction confidence is below a threshold of 0.4, none of the four classes are predicted and instead the label *Unknown* is cast. Both models classified all 4,450 images. SWIN-Base classified 83 images as *Unknown* (1.865%) while DeiT-Base classified 181 images as *Unknown* (4.067%). The other classes also differ in their proportions. SWIN-Base classifies almost half of all images as *Neutral* (n = 2,055; 46.180%) but only 6.494% as *Pessimistic* (n = 289). DeiT-Base on the other hand classifies *Pessimistic* more often (n = 454; 10.202%) but *Neutral* less often (n = 1,780; 40.000%).

4.4 Exploratory Analysis

Figure 5

Normalized confusion matrix for SWIN-Base and DeiT-Base predictions



Note. N = 4,450. Integers show raw counts, percentages show row proportions. *Unknown* label is cast if prediction confidence < 0.4. Own figure.

To test for differences in the two prediction sets we conduct Bowker's test of symmetry for paired categorical data across five classes – our four prediction classes plus the *Unknown* class – and also calculate the strength of association. Results indicate a statistically significant difference in prediction distributions between SWIN-Base and DeiT-Base (Bowker-$\chi^2(10) = 165.166$, $p < .001$). However, the strength of association is negligible (Cramer-V = .096). Differences in model predictions are visualized in Figure 5. It is important to note that this is not an indicator for poor model performance. We already measured model performance individually in section 4.1 using metrics like F1 and Cohen's Kappa.

5 Discussion

5.1 Conclusion

In this paper we present two vision transformer models for emotion recognition in visual imagery and demonstrate their application to climate change communication. Our primary contribution is practical: we provide social scientists with what we believe to be among the first well-documented, fully open-source transformer models for this task. By releasing all code, training data sources, and model checkpoints, our approach is completely reproducible without requiring expensive cloud infrastructure. Answering *RQ1*, we show that training and running these models is feasible on local workstations and even everyday mobile devices such as an Apple MacBook Air, removing a significant practical barrier to the adoption of advanced image classification tools in social science research.

Our results demonstrate a performance advantage of SWIN-Base over DeiT-Base, both in general transfer learning ($\Delta F1 = .025$) and especially after adaptation to real-world data ($\Delta F1 = .160$). We attribute this to a fundamental difference in how the two models process images. DeiT-Base analyses all parts of an image simultaneously and at equal resolution, which works well with large amounts of training data but seems to generalise less reliably when data is limited. SWIN-Base, by contrast, first focuses on small local regions before gradually building up a broader understanding of the whole scene, which is essentially how the human eye works (Liu et al., 2021). This makes it better suited to situations where training data is limited and the images to be classified look different from the training material, as is the case in domain adaptation to web-scraped social media imagery. Overall, we find some support for *RQ2*, as SWIN-Base delivers strong results on both datasets, even though DeiT-Base remains below expectations.

This finding also speaks on earlier research on context-aware emotion recognition. Lee et al. (2019) who published the CAER-S dataset used in this paper, simultaneously introduced an emotion classifier that combines facial expression recognition with scene context. Because their work predates the vision transformer, which was introduced in 2021, they achieved this task by running two CNNs in parallel. Our results suggest that a single transformer model can accomplish the same goal more effectively since self-attention allows transformers to analyse faces, body posture, objects and background scenery within one unified process instead of two separate components. However, a direct comparison between the CaerNet model by Lee et al. (2019) and our implementation is necessary to accurately judge relative performance. We leave this comparison to future research.

Although our models were developed in the context of climate change communication research, the four-class taxonomy with *Optimistic*, *Pessimistic*, *Hostile* and *Neutral* is relevant across a wide range of social science applications, including political communication, health campaigns, and crisis communication research. The domain adaptation results for SWIN-Base ($\kappa = .684$) suggest that researchers working in related fields can customize our open-source checkpoints to their own image corpora with relatively modest annotation effort (N = 410 in our case). Given that these images were web-scraped from Instagram, rather then collected under controlled, standardized conditions, this result indicates the approach is robust to confusing and unorganized real-world data social scientists are likely to encounter. Researchers who want to use our models can do so within ten lines of code using the simplified inference script published on our Huggingface repository.

5.2 Limitations

Our work is limited in several aspects. First of all, we only classify four emotions. This decision is grounded in climate change communication research (O'Neill & Nicholson-Cole, 2009; O'Neill, 2013; Chapman et al., 2017; Nabi et al., 2018; Brosch, 2021; Dezma et al., 2024), which identifies those four emotions as most relevant. However, emotion taxonomies proposed by Plutchik (1980), Ekman (1992) and Cao et al. (2014) contain a larger, more fine-grained number of emotions. A larger number of classes can increase training difficulty substantially, requiring much larger training corpora than the 60,000 images we use. Even then, more subtle emotions might be misclassified more often, hurting model performance and robustness. On a similar matter, our models cast single hard labels for each classification. However, emotion is a complex psychological phenomenon and not strictly discrete. Several emotions can be present at the same time. Furthermore, research in psychology and communication science indicates promising results in mapping emotions to a two-dimensional space using the concepts of valence and arousal (Russell, 1980; Russell & Barrett, 1999; Barrett, 2006; Barrett, 2017). Those concepts argue that discrete emotions like joy or anger are only interpretations of a continuous neuropsychological state. Our models are unable to accommodate this approach. Training a model to place emotion in continuous 2D-spaces is a very challenging task requiring huge training corpora. Moreover, establishing a solid ground truth on continuous labels would either require physiological measurements or complex annotation processes substantially beyond standard categorical annotations.

Finally, we have not systematically evaluated our models for bias. We have a selection bias by using the CAER-S dataset which uses only images from films and TV-series. Lee et al. (2019) do not specify which films and series the dataset consists of. It is therefore possible that scenes containing individuals underrepresented in the training data are more often misclassified. Underrepresented groups might contain but are not limited to age, gender and ethnicity. The same is true for our domain adaptation dataset which consists of images scraped from Instagram and related to climate change activism. These images probably reflect communication strategies from this specific group and model checkpoints might not be applicable to other visual domains. This limitation is somewhat mitigated by the fact that we publish both the checkpoints from general transfer learning and domain adaptation. We recommend building on the checkpoints from general transfer learning which are not domain adapted to eliminate this specific form of bias. A thorough evaluation on model performance across diverse subgroups is a task we leave for future work. We encourage researchers to audit our models and improve them where necessary.

5.3 Lookout

Our models provide researchers with powerful image classification tools. However, much remains to be done, given that machine learning research is among the fastest developing fields today. We do not compare our approach to a more traditional CNN approach. While our models are computationally cheap in the context of vision transformers they are still more complex than CNNs. It is important to compare our approach to more baseline approaches and establish differences in computational cost and prediction performance allowing to assess performance-efficiency tradeoffs between architectures.

Furthermore, models using multidimensional labels or soft labels are intriguing, both from a practical and a theoretical perspective. Emotion is often complex and not unidimensional. Multiple emotions might occur at the same time. Our models are unable to accommodate for this aspect and can only cast a single hard label per scene. We use soft labels during training to improve generalization in a process called MixUp. However, predicting soft labels in inference is a very challenging task. Given that even with four classes we need a sophisticated training setup it appears non-trivial to improve training performance far enough to allow for the classification of soft labels. Our research indicates that hierarchical self-attention mechanisms are superior to more traditional transformer global context mechanisms. Future research could explore different implementations of hierarchical mechanisms to improve model performance.

Another promising approach in deep learning is called *world models* (Schmidhuber, 1990; Ha & Schmidhuber, 2018; Hafner et al., 2019). World models build an internal spatial and temporal representation of an environment, for example the real world. They can then predict changes over time with regard to actions by different actors. Today, world models are predominantly used in robotics, autonomous driving and for advanced simulations in physics and engineering (Hansen et al., 2023; Wu et al., 2023; Nakamura et al., 2023; Yin et al., 2026). However, by trying to predict events in the real world using true contextual and causal understanding they have enormous potential for use in social science. We are not aware of any existing world model for visual emotion recognition. Creating a world model for this task will be a major advancement for the field.

References

Aminoff, E. M., Kveraga, K., & Bar, M. (2013). The role of the parahippocampal cortex in cognition. *Trends in Cognitive Sciences*, *17*(8), 379–390. https://doi.org/10.1016/j.tics.2013.06.009

Barrett, L. F. (2006). Solving the Emotion Paradox: Categorization and the Experience of Emotion. *Personality and Social Psychology Review*, *10*(1), 20–46. https://doi.org/10.1207/s15327957pspr1001_2

Barrett, L. F. (2017). *How emotions are made: The secret life of the brain*. Mariner Books.

Barrett, L. F., Mesquita, B., & Gendron, M. (2011). Context in Emotion Perception. *Current Directions in Psychological Science*, *20*(5), 286–290. https://doi.org/10.1177/0963721411422522

Brosch, T. (2021). Affect and emotions as drivers of climate change perception and action: A review. *Current Opinion in Behavioral Sciences*, *42*, 15–21. https://doi.org/10.1016/j.cobeha.2021.02.001

Buck, I., Foley, T., Horn, D., Sugerman, J., Fatahalian, K., Houston, M., & Hanrahan, P. (2004). *Brook for GPUs: Stream Computing on Graphics Hardware*. https://graphics.stanford.edu/papers/brookgpu/

Bulwer, J. (1649). *Pathomyotamia, or, A dissection of the significative muscles of the affections of the minde being an essay to a new method of observing the most important movings of the muscles of the head, as they are the neerest and immediate organs of the voluntarie or impetuous motions of the mind: With the proposall of a new nomenclature of the muscles / by J.B., sirnamed the Chirosopher.* W.W. for Humphrey Moseley.

Canal, F. Z., Müller, T. R., Matias, J. C., Scotton, G. G., De Sa Junior, A. R., Pozzebon, E., & Sobieranski, A. C. (2022). A survey on facial emotion recognition techniques: A state-of-the-art literature review. *Information Sciences*, *582*, 593–617. https://doi.org/10.1016/j.ins.2021.10.005

Cao, H., Cooper, D. G., Keutmann, M. K., Gur, R. C., Nenkova, A., & Verma, R. (2014). CREMA-D: Crowd-Sourced Emotional Multimodal Actors Dataset. *IEEE Transactions on Affective Computing*, *5*(4), 377–390. https://doi.org/10.1109/TAFFC.2014.2336244

Chapman, D. A., Lickel, B., & Markowitz, E. M. (2017). Reassessing emotion in climate change communication. *Nature Climate Change*, *7*(12), 850–852. https://doi.org/10.1038/s41558-017-0021-9

Corneanu, C. A., Simon, M. O., Cohn, J. F., & Guerrero, S. E. (2016). Survey on RGB, 3D, Thermal, and Multimodal Approaches for Facial Expression Recognition: History, Trends, and Affect-Related Applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *38*(8), 1548–1568. https://doi.org/10.1109/TPAMI.2016.2515606

da Vinci, L. (1651). *Trattato della Pittura*.

Dézma, L., Bigazzi, S., Sarrica, M., Siegler, A., Serdült, S., & Rizzoli, V. (2025). Social Representation of Global Climate Change: An Exploratory Study Focusing on Emotions. *Journal of Constructivist Psychology*, *38*(2), 328–349. https://doi.org/10.1080/10720537.2024.2310828

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale* (arXiv:2010.11929). arXiv. https://doi.org/10.48550/arXiv.2010.11929

Duong, P. A., Luong, C., Bommana, D., & Jiang, T. (2025). *CHEER-Ekman: Fine-grained Embodied Emotion Classification* (Version 2). arXiv. https://doi.org/10.48550/ARXIV.2506.01047

Ekman, P. (1992). An argument for basic emotions. *Cognition and Emotion*, *6*(3–4), 169–200. https://doi.org/10.1080/02699939208411068

Fukushima, K. (1980). Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biological Cybernetics*, *36*(4), 193–202. https://doi.org/10.1007/BF00344251

Gao, C., Ajith, S., & Peelen, M. V. (2025). Object representations drive emotion schemas across a large and diverse set of daily-life scenes. *Communications Biology*, *8*(1), 697. https://doi.org/10.1038/s42003-025-08145-1
Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. The MIT press.
Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). *Challenges in Representation Learning: A report on three machine learning contests* (Version 1). arXiv. https://doi.org/10.48550/ARXIV.1307.0414
Ha, D., & Schmidhuber, J. (2018). *World Models*. https://doi.org/10.48550/ARXIV.1803.10122
Hafner, D., Lillicrap, T., Ba, J., & Norouzi, M. (2019). *Dream to Control: Learning Behaviors by Latent Imagination* (Version 3). arXiv. https://doi.org/10.48550/ARXIV.1912.01603
Haim, M., Haßler, J., Lübke, S., Ozornina, N., Stengl, L., Geigl, M., Plank, B., Peng, S. L., Zhou, S., Ommer, B., Schusterbauer, J., Grabe, M. E., & Mohammad, S. (2026). *KLIMAMEMES dataset* [Computer software]. https://klimamemes.ifkw.lmu.de/index.php/about-the-project/
Hansen, N., Su, H., & Wang, X. (2023). *TD-MPC2: Scalable, Robust World Models for Continuous Control* (Version 2). arXiv. https://doi.org/10.48550/ARXIV.2310.16828
He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90
Huggingface. (2026). *Huggingface Models* [Repository]. https://huggingface.co/models
Ibrahim, E., Ezzat Ghoraba, M., & Ezzat Ghoraba, A. (2026). Multimodal emotion recognition using hybrid deep feature fusion under speaker-independent evaluation. *Scientific Reports*, *16*(1), 19584. https://doi.org/10.1038/s41598-026-58836-w
Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *Proceedings of the 26th International Conference on Neural Information Processing Systems*, *1*, 1097–1105. https://doi.org/10.5555/2999134.2999257
Landis, R. J., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, *33*(1), 159–174.
Lecun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278–2324. https://doi.org/10.1109/5.726791
Lee, J., Kim, S., Kim, S., Park, J., & Sohn, K. (2019). Context aware emotion recognition networks. *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). *Swin Transformer: Hierarchical Vision Transformer using Shifted Windows*. arXiv. https://doi.org/10.48550/arXiv.2103.14030
Luebke, S. M., Ozornina, N., Haim, M., & Haßler, J. (2025). *Codebook: Climate Change and Social Media*. https://osf.io/82khj/files/adkzg
Mocanu, B., Tapu, R., & Zaharia, T. (2023). Multimodal emotion recognition using cross modal audio-video fusion with attention and deep metric learning. *Image and Vision Computing*, *133*, 104676. https://doi.org/10.1016/j.imavis.2023.104676
Mollahosseini, A., Hasani, B., & Mahoor, M. H. (2017). *AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild*. https://doi.org/10.48550/ARXIV.1708.03985
Moser, S. C. (2016). Reflections on climate change communication research and practice in the second decade of the 21st century: What more is there to say? *WIREs Climate Change*, *7*(3), 345–369. https://doi.org/10.1002/wcc.403
Nabi, R. L., Gustafson, A., & Jensen, R. (2018). Framing Climate Change: Exploring the Role of Emotion in Generating Advocacy Behavior. *Science Communication*, *40*(4), 442–468. https://doi.org/10.1177/1075547018776019

Nakamura, K., Peters, L., & Bajcsy, A. (2025, June 21). Generalizing Safety Beyond Collision-Avoidance via Latent-Space Reachability Analysis. *Robotics: Science and Systems XXI*. Robotics: Science and Systems 2025. https://doi.org/10.15607/RSS.2025.XXI.113

O'Neill, S. J. (2013). Image matters: Climate change imagery in US, UK and Australian newspapers. *Geoforum*, *49*, 10–19. https://doi.org/10.1016/j.geoforum.2013.04.030

O'Neill, S., & Nicholson-Cole, S. (2009). "Fear Won't Do It": Promoting Positive Engagement With Climate Change Through Visual and Iconic Representations. *Science Communication*, *30*(3), 355–379. https://doi.org/10.1177/1075547008329201

Pereira, R., Mendes, C., Ribeiro, J., Ribeiro, R., Miragaia, R., Rodrigues, N., Costa, N., & Pereira, A. (2024). Systematic Review of Emotion Detection with Computer Vision and Deep Learning. *Sensors*, *24*(11), 3484. https://doi.org/10.3390/s24113484

Plutchik, R. (Ed.). (1984). *Emotion: Theory, research, and experience. Vol. 1: Theories of emotion* (2. Druck). Acad. Press.

Rahman, R. R., Choi, H. Y., Lim, J., Lee, G. E., Lee, S., Cho, C., & Damevski, K. (2026). Toward Generalized Emotion Recognition in VR by Bridging Natural and Acted Facial Expressions. *Sensors*, *26*(3), 845. https://doi.org/10.3390/s26030845

Ricci, M. M. G., & Levi-Civita, T. (1900). Methodes de calcul differentiel absolu et leurs applications. *Mathematische Annalen*, *54*(1–2), 125–201. https://doi.org/10.1007/BF01454201

Riemann, B. (1868). *Ueber die Hypothesen, welche der Geometrie zu Grunde liegen. [Habilitationsschrift von 1854]*.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, *39*(6), 1161–1178. https://doi.org/10.1037/h0077714

Russell, J. A. (1994). Is there universal recognition of emotion from facial expression? A review of the cross-cultural studies. *Psychological Bulletin*, *115*(1), 102–141. https://doi.org/10.1037/0033-2909.115.1.102

Russell, J. A., & Barrett, L. F. (1999). Core affect, prototypical emotional episodes, and other things called emotion: Dissecting the elephant. *Journal of Personality and Social Psychology*, *76*(5), 805–819. https://doi.org/10.1037/0022-3514.76.5.805

Schmidhuber, J. (1990). *Making the world differentiable: On using fully recurrent self-supervised neural networks for dynamic reinforcement learning and planning in non-stationary environments [Technical Report FKI-126-90, Technische Universität München (TUM)]* [Technical Report]. Technische Universität München.

Shan, C., Gong, S., & McOwan, P. W. (2009). Facial expression recognition based on Local Binary Patterns: A comprehensive study. *Image and Vision Computing*, *27*(6), 803–816. https://doi.org/10.1016/j.imavis.2008.08.005

Simonyan, K., & Zisserman, A. (2014). *Very Deep Convolutional Networks for Large-Scale Image Recognition* (Version 6). arXiv. https://doi.org/10.48550/ARXIV.1409.1556

Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., & Jegou, H. (2021). *Training data-efficient image transformers & distillation through attention*. arXiv. https://doi.org/10.48550/arXiv.2012.12877

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). *Attention Is All You Need* (Version 7). arXiv. https://doi.org/10.48550/ARXIV.1706.03762

Wu, P., Escontrela, A., Hafner, D., Abbeel, P., & Goldberg, K. (2023). Daydreamer: World models for physical robot learning. *Proceedings of Machine Learning Research*, 2226–2240.

Wu, Y., Mi, Q., & Gao, T. (2025). A Comprehensive Review of Multimodal Emotion Recognition: Techniques, Challenges, and Future Directions. *Biomimetics*, *10*(7), 418. https://doi.org/10.3390/biomimetics10070418

Yin, T., Mei, Z., Zheng, Z., Yamane, M., Wang, D., Sceats, J., Bateman, S. M., Zha, L., Badithela, A., Shorinwa, O., & Majumdar, A. (2026). *PlayWorld: Learning Robot World Models from Autonomous Play* (Version 3). arXiv. https://doi.org/10.48550/ARXIV.2603.09030
Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2017). *mixup: Beyond Empirical Risk Minimization* (Version 2). arXiv. https://doi.org/10.48550/ARXIV.1710.09412

Appendix

Glossary

Note that his glossary is written with social scientists in mind. We cannot explain all the concepts appearing in this paper in their full depth as that would require advanced knowledge in mathematics and computer science. We therefore focus on simple, basic explanations.

Attention mechanism

A machine learning technique that enables a model to focus on the most relevant parts of the input while making a prediction. In Vision Transformers, attention determines which image regions are most important for interpreting each image patch.

Backbone (model)

The main part of a neural network that learns useful patterns from images. In transfer learning, the backbone is usually pre-trained on a large image dataset and then adapted to a new task.

Batch

A small group of images processed together during training instead of processing the entire dataset at once.

Batch size

The number of images contained in one batch during training.

BF16 (Brain Floating Point 16)

A numerical format that allows modern hardware to perform calculations faster while using less memory than standard 32-bit precision, with little effect on model accuracy.

Classifier head

The final part of a neural network that assigns an input image to one of the predefined classes (e.g., optimistic, pessimistic, hostile or neutral).

Convolution

A mathematical operation used in convolutional neural networks to detect simple visual patterns such as edges, shapes or textures.

Convolutional Neural Network (CNN)

A type of deep learning model that analyses images by detecting increasingly complex visual features. CNNs have been the dominant approach in computer vision for many years.

Cross Entropy Loss

A commonly used measure of how different the model's predictions are from the correct labels. Smaller values indicate more accurate predictions.

Data augmentation

Artificially creating additional training samples by applying small modifications such as flipping, rotating or cropping to existing images. This helps improve the model's ability to generalise.

Domain adaptation

Adapting a model that was trained on one type of images so that it performs well on a different but related type of images.

Embedding space

A numerical representation in which visually similar images are located close to one another and visually dissimilar images located far apart from each other.

Encoder block

The basic processing unit of a Vision Transformer. Each encoder block analyses the relationships between image patches and learns increasingly meaningful visual representations.

Epoch

One complete pass through the entire training dataset.

F1-score

A performance measure that combines precision and recall into a single value. Higher values indicate better classification performance, particularly when classes are unevenly distributed.

Feature extraction

The process of identifying meaningful visual characteristics, such as objects, facial expressions or colours, that help distinguish between image classes.

Fine-tuning

Continuing the training of a pre-trained model on a new dataset so that it becomes specialised for a specific task.

Floating-point precision

The numerical accuracy used for calculations during model training and inference. In simplified terms how many decimals are used for mathematical computations. Lower precision formats are often faster and require less memory but introduce larger rounding errors.

Global context

Information from the entire image rather than only a small region. Vision Transformers can use global context to better understand how different parts of an image relate to one another.

Gradient scaling

A technique used during mixed-precision training that reduces numerical errors and helps ensure stable model training. Only needed in mixed-precision training.

Hierarchical self-attention

A method used by some Vision Transformers that gradually combines information from small image regions into larger ones, allowing the model to recognise both fine details and the overall scene. The model starts by evaluating small patches of an image which are then merged to larger and larger areas until the entire image is combined to one large region.

ImageNet

A large collection of more than one million labelled images that is commonly used to pre-train computer vision models before they are adapted to other tasks.

Inference

Using a trained model to make predictions for new, previously unseen images.

Kernel (CNN)

A small filter that moves across an image to detect specific visual patterns, such as edges or textures.

Learning rate

A training setting that determines how much the model changes after each learning step. A higher learning rate allows the model to make bigger steps but can introduce instability. A lower learning rate limits the model steps and can lead to suboptimal learning progress.

Loss

A numerical measure of prediction error during training. The objective of training is to minimise the loss.

Mixed-precision training

A training technique that performs some calculations with lower numerical precision (e.g. BF16) and others with higher precision (e.g. FP32). This reduces memory usage and speeds up training while maintaining reliable results.

Mixup

A data augmentation technique that creates new training images by combining pairs of images and their labels. This can improve the model's ability to generalise.

Multi-head self-attention

A mechanism that allows a Vision Transformer to examine multiple relationships between different parts of an image simultaneously. This enables the model to capture different aspects of the image at the same time.

Optimiser

An algorithm that updates the model after each training step in order to reduce prediction errors.

Overfitting

A problem in which a model learns the training data too closely and therefore performs poorly on new, unseen data. Basically, the model does not actually learn anymore and instead just “remembers” the images it has already seen.

Patch (Vision Transformer)

A small square section of an image. Vision Transformers divide an image into patches and analyse these instead of the complete image at once.

Patch embedding

The process of converting each image patch into a numerical representation that can be processed by a Vision Transformer. Vision Transformers work with mathematical operations and therefore need to translate an image into numbers.

Pooling

A technique used in convolutional neural networks to reduce the size of image representations while preserving the most important information.

Positional embedding

Additional information that tells a Vision Transformer where each image patch is located within the original image. Transformers have no inherent understanding of dimensionality, therefore the patch embedding is needed to tell the model where a specific input is located.

Pre-training

The initial training of a model on a large, general-purpose dataset before it is adapted to a specific task. Pre-training allows the model to learn broadly useful visual representations. Pre-training is often conducted by large corporations like Google, Microsoft or Meta. Individual researchers rarely conduct pre-training as it is very expensive and requires a lot of specialised computer hardware.

Precision (classification)

The proportion of images predicted to belong to a class that are actually members of that class. Precision is one of the most used metrics in machine learning along accuracy, recall and F1.

Precision (floating point)

The numerical accuracy used for calculations in a computer. Higher precision generally improves accuracy but requires more memory and computing power.

Recall

The proportion of all images belonging to a class that are correctly identified by the model. Recall is one of the most used metrics in machine learning along accuracy, precision and F1

Self-attention

An attention mechanism that allows a Vision Transformer to determine which parts of an image are most relevant when analysing each image patch.

Shifted-window mechanism

A technique used for example in SWIN Transformers in which neighbouring image regions are analysed together in alternating processing steps. This enables the model to exchange information between neighbouring image regions while remaining computationally efficient. Basically, not the entire image is analysed simultaneously but instead split into many regions which are analysed individually. Then, regions are partially overlapped and analysed again, making comparisons between the regions possible. This requires less mathematical operations than analysing everything at the same time.

Tensor

A structured collection of numbers used to represent images and other data inside a neural network. Images are stored as tensors while they are processed by the model. A tensor in data science can be imagined as a matrix. This matrix can have any number of dimensions.

Test split

The portion of the dataset that is used only after training has finished to evaluate how well the model performs on previously unseen data.

Training

The process of teaching a neural network by repeatedly comparing its predictions with the correct answers and gradually improving its performance.

Training split

The portion of the dataset that is used to train the model by allowing it to learn from labelled examples.

Transfer learning

A method in which a model that has already learned from a large dataset is adapted to a new task using a smaller, task-specific dataset.

Validation split

The portion of the dataset used during training to monitor model performance and to guide decisions such as selecting hyperparameters or determining when training should stop.

Vision Transformer (ViT)

A deep learning model that analyses images by dividing them into small patches and learning relationships between these patches using self-attention rather than convolution.

Github

Our entire code and key visualizations are published in a public Github repository under this URL: https://github.com/47867/ViT-ER. Code and visualizations are distributed "as is" under a MIT license.

Huggingface

Model checkpoints after general transfer learning as well as domain adaptation are published in a public Huggingface repository under these URLs:
SWIN: https://huggingface.co/Simon14142/emotion-recognition-4class-swin
DeiT: https://huggingface.co/Simon14142/emotion-recognition-4class-deit
Model checkpoints are distributed under an Apache-2.0 license.

SHA256 checksums are provided for for all model checkpoints for file integrity verification.

File	SHA256 checksum
model.safetensors (SWIN)	d7057ef18959183bb0cdada1182429753a567e047 5bdfde1e2dae7cfac4ed939
model.safetensors (DeiT)	0ccd50ab34b3518f2bdc3d3847885c7b3e58ea593 0025ce3a14b61633e37678d
adaptation_stage-2_SWIN- Base_checkpoint_final.safetensors	4cad91a85be40239084a9e86e41526a9eb91f13ee 2a9c90b80db57f7a1babb79
adaptation_stage-2_DeiT- Base_checkpoint_final.safetensors	67b7c1cc9102391b33de58107d3dd36d0ca19892 26f39ec7cad4f892c0d93484

Additionally, a simplified inference script is published called *emotion-classifier.py*. It allows to setup the model for single-image classification in less than ten lines of code. Explanations are provided in the repository description. Requires *torch, timm, safetensors, huggingface_hub, pillow* and *torchvision* libraries plus dependencies in Python. GPU-acceleration is recommended. The inference script is also published under an Apache-2.0 license.

Codebook

Our codebook contains only the four emotion classes. Technical parameters like ID or image-path were not coded and instead automatically cast. ID was cast by pandas dataframe indexing and image-path was cast by our chosen Linux KDE file management system, called Dolphin.

All classes are coded dichotomous either with 1 (= does exist) or 0 (= does not exist).

Class descriptions are directly adapted from Luebke et al. (2025, p. 75 - 78) who published the codebook for the KLIMAMEMES dataset.

Emotion	Description
Optimistic	If joy is recognizable in an image. Joy can be: - laughing or smiling - relaxed facial expressions - open body posture - relaxed body posture - activity like „jumping with joy“ - visual joyful exclamations
Pessimistic	If either sadness or fear or both is recognizable in an image. Fear can be: - eyes wide open - clenched teeth - tense facial features - looking away or at hands - visual tendency to flee - trembling or twitching - insecure, tense smiling

	- tense or crouching posture Sadness can be: - crying, wailing - visible tears - hands in front of the eyes - downturned corners of the mouth - lowered or downward gaze - head hanging - slumped shoulders, collapsed posture - self-comforting gestures, embracing oneself
Hostile	If either anger or disgust or both is recognizable in an image. Anger can be: - compressed lips - fixed, hard gaze - flaring of the nostrils - frown - contracted eyebrows - shouting / yelling - raised arms, clenched fists - aggressive body posture Disgust can be: - wrinkled nose - raised upper lip, curled upper lip - retching, gagging - squinting, narrowed eyes - body leaning away from an object or person - hand covering the mouth - visible expression of revulsion
Neutral (Baseline)	If no apparant emotion of the above is recognizable in an image.

Hardware specifications

Hardware	Specification
GPU	NVIDIA Blackwell GB203 16 GB GDDR7 VRAM 28 Gigabit/s over 256-bit bus = 896 GB/s (~ 834 GiB/s) 300 watt power target (factory default) connected directly to CPU via PCI-Express Gen. 5.0 with 16 lanes
CPU	Intel 12th Gen. (Alder Lake) Core i5-12600K 10 cores (6 performance cores / 4 efficiency cores)

	16 threads 4.90 GHz boost clock (factory default) 125 watt power target (factory default)
CPU Chipset	Z690 for LGA-1700
DRAM	64 GB (4 x 16 GB) Dual channel operation Running at JEDEC specification DDR4 -3200MT/s CL: 16 tRCD: 20 tRP: 20 tRAS: 38 1.35 volts Manufactured by Micron
Mass storage	1 TB (~ 928.2 GiB) SAMSUNG 870QVO ssd SATA-III / SATA-600 over CPU chipset

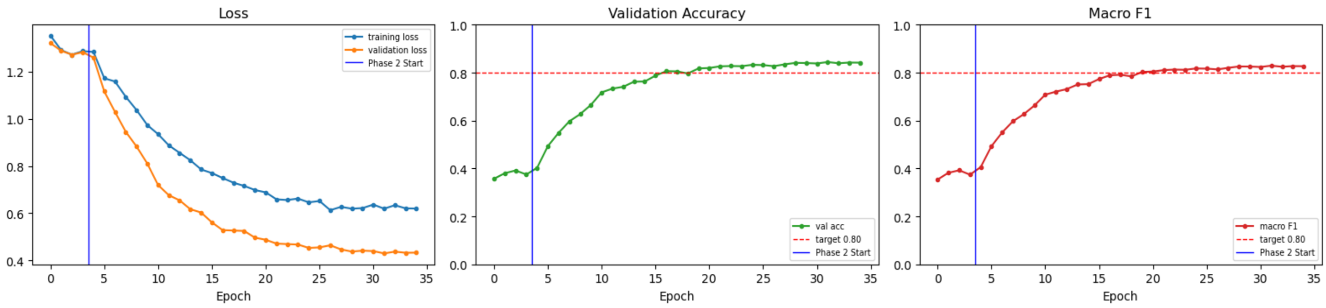
Software specifications

Software	Specification
Distribution	Linux Fedora 43 for Workstation
Kernel	Linux 7.0.12-101.fc43.x86_64
NVIDIA CUDA	v12.8
NVIDIA driver	v580.159.04
Python	v3.13.13
Conda	v26.1.1
Jupyter Lab	v4.5.6

Figures

Figure 6

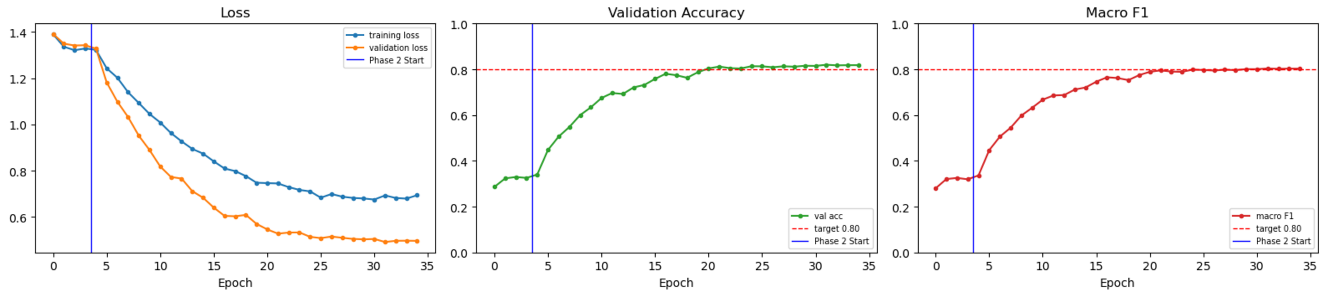
Training curves for SWIN-Base transfer learning



Note. Model trained for 32 epochs. Batch-size = 64.

Figure 7

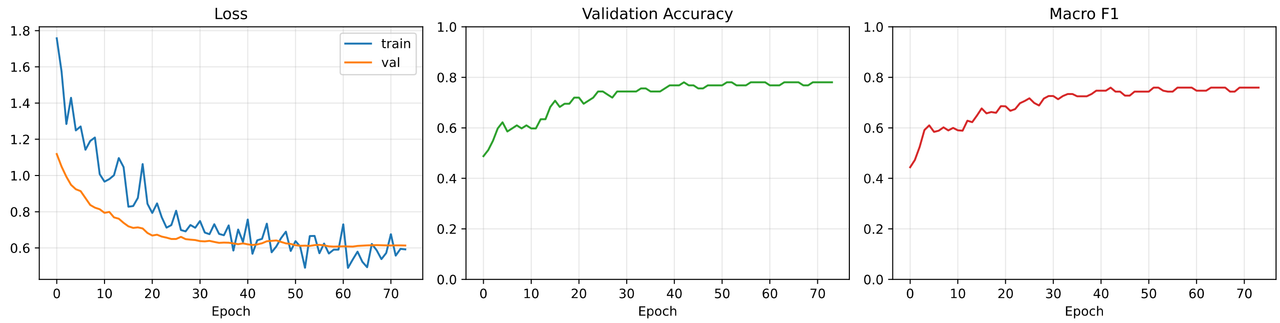
Training curves for DeiT-Base transfer learning



Note. Model trained for 32 epochs. Batch-size = 64.

Figure 8

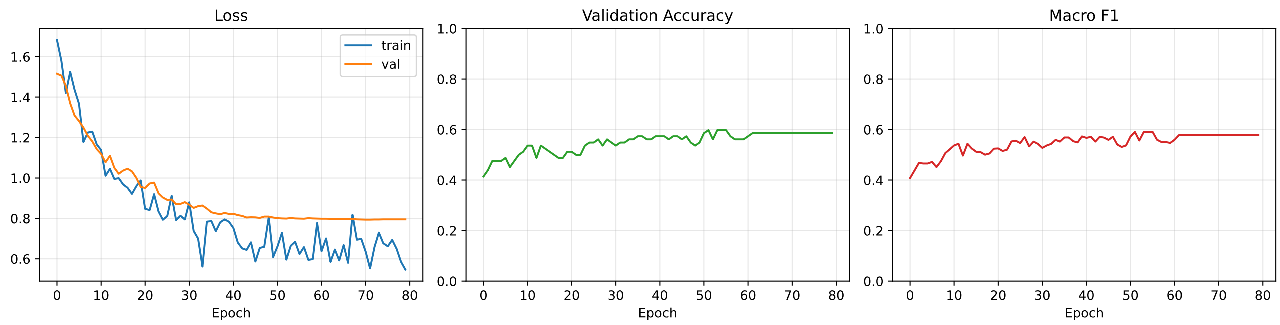
Training curves for SWIN-Base domain adaptation



Note. Model trained for 74 epochs and then stopped early as no new global minimum for validation loss was reached for 15 consecutive epochs. Batch-size = 16. Spikes in train loss due to small batch size.

Figure 9

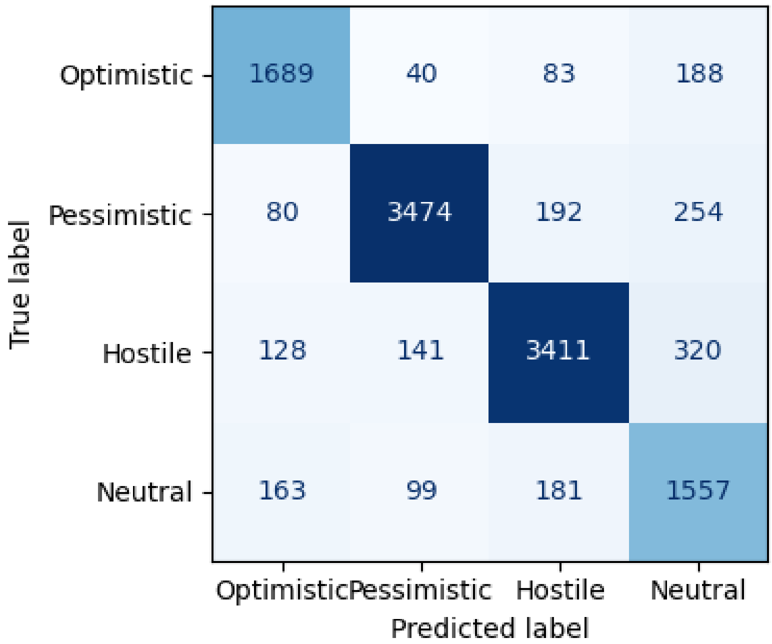
Training curves for DeiT-Base domain adaptation



Note. Model trained for 80 epochs. Batch-size = 16. Spikes in train loss due to small batch size.

Figure 10

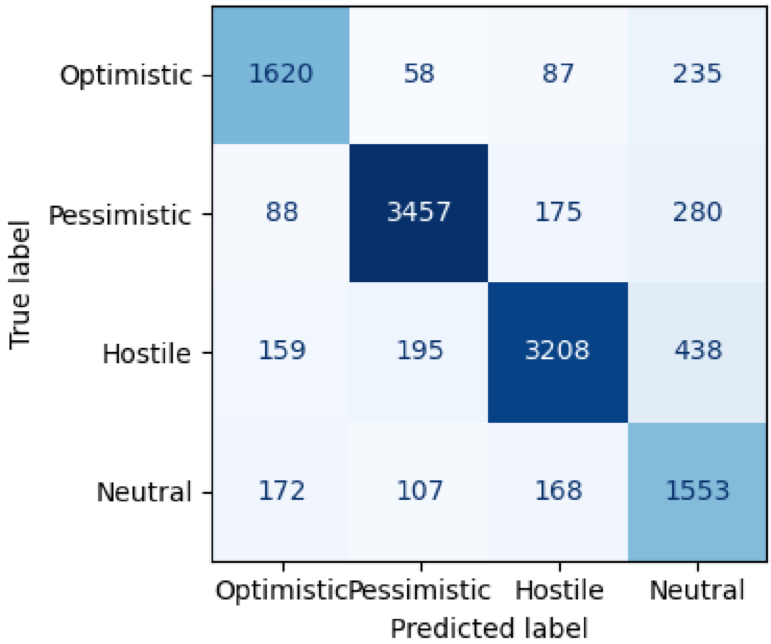
Confusion matrix for SWIN-Base transfer learning



Note. N = 12,000.

Figure 11

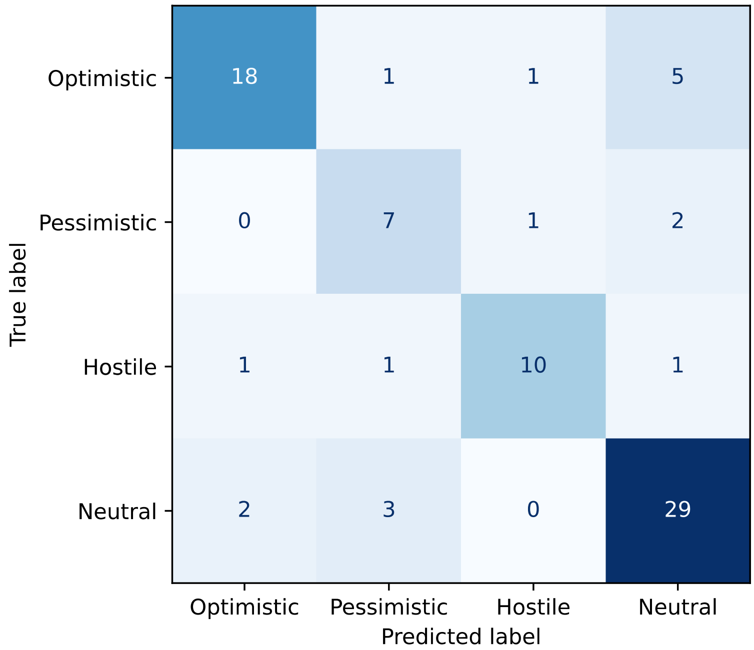
Confusion matrix for DeiT-Base transfer learning



Note. N = 12,000.

Figure 12

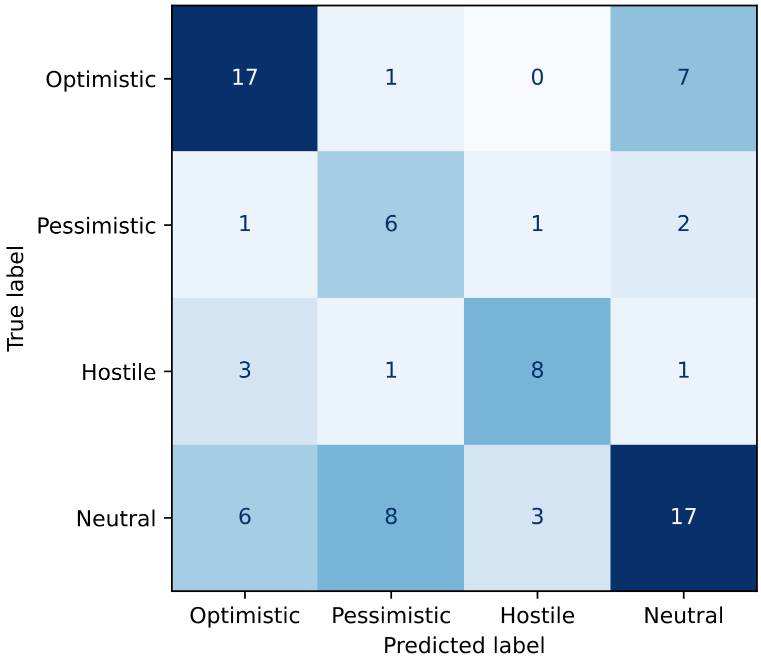
Confusion matrix for SWIN-Base domain adaptation



Note. N = 82.

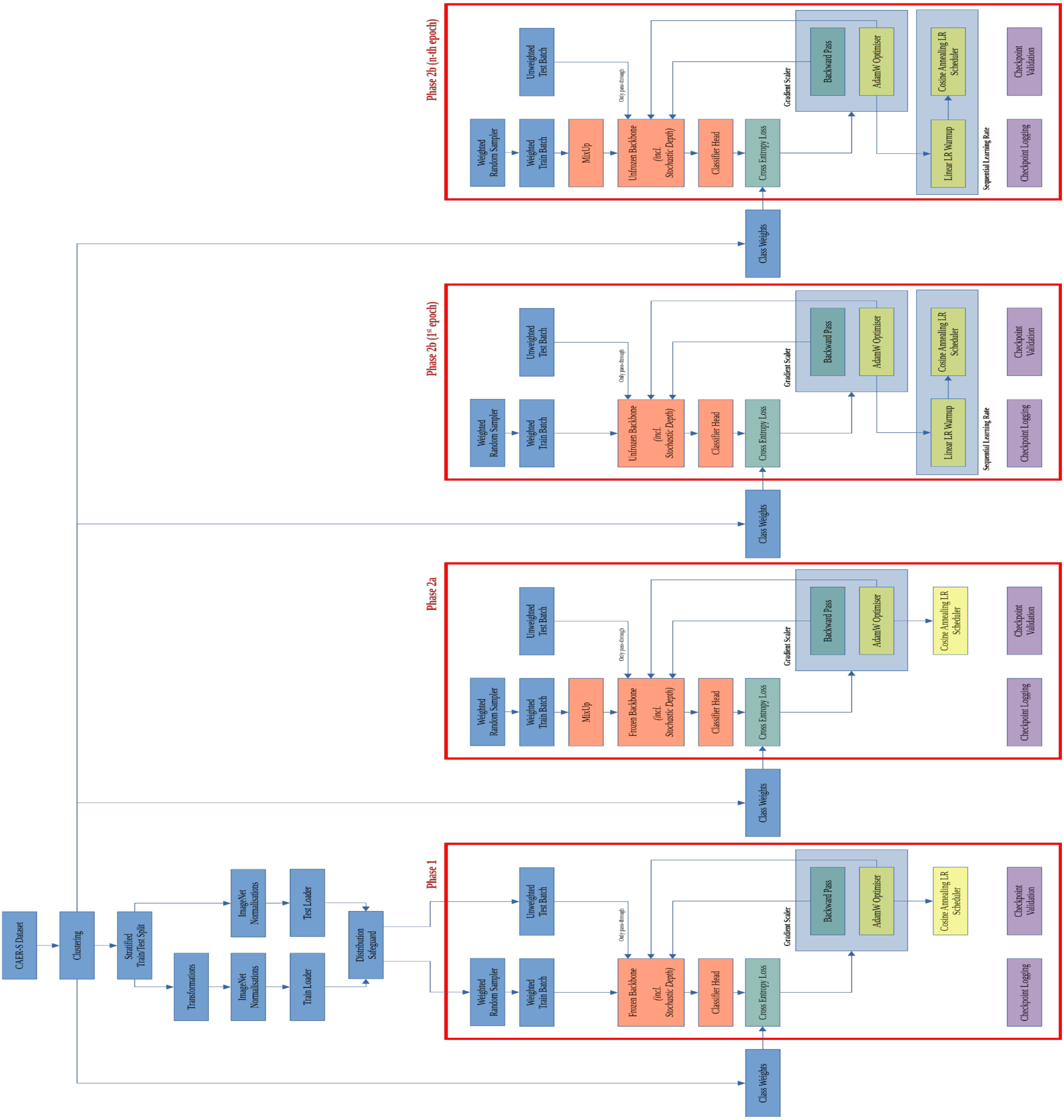
Figure 13

Confusion matrix for DeiT-Base domain adaptation



Note. N = 82

Visualization of model architecture



Note. Colour codings: Blue = Data, Orange = Model, Green = Loss computation, Yellow = Optimiser and learning rate, Purple = Validation, Red = Epoch blocks.